

AE 3610

Evaluating Cross-Sectional Designs for Aerospace Structural Integrity

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Abstract

This report investigates the structural performance of specimens with different internal cross-sectional patterns that were subjected to axial compression loading. The three main internal designs tested were horizontal bar patterns, vertical patterned columns, and a patterned X-shaped structure. SolidWorks was used to generate 3D models of each beam, which were 3D printed using PLA filament material. Critical dimensions such as thickness and exterior shape were kept consistent across each specimen to ensure only the internal structure differed in each beam. Each specimen was compressed to their failure points using an Instron load frame, which recorded the load applied and frame displacement. Key data, including force-displacement and stress-strain relationships, were analyzed to compare the designs based on their axial stiffness, yield stress, and fracture loads. Results indicate that horizontal bar specimens exhibited the highest structural integrity, while vertical bar specimens demonstrated early buckling. X-shaped reinforcement designs offered intermediate performance but showed potential for optimization. These findings provide insights into improving beam cross-section designs for enhanced load-bearing capacity in aeronautical applications.

Introduction

Aircraft structures are made up of subcomponents designed to withstand aerodynamic forces and maximize lifespan. Among these, spars play a critical role, providing static stability to wings and fuselages. The cross-sectional configurations of spars are optimized to endure compressive and tensile loads in various directions. The following report explores three different types of spar cross-sections to analyze how their structures vary their ability to carry loads. These designs include X-shaped reinforcements, longitudinal bars, and transverse bars.

The designs were first modeled in SolidWorks and then 3D printed using Polylactic Acid (PLA) filament. Axial compression loading experiments were conducted on each type of spar in a structural testing facility using an Instron tensile tester. Each spar was loaded until fracture while measuring the load applied and corresponding axial displacement over the experiment time interval. The dimensions of the spars were measured with calipers so that the force-displacement relation information can be used to create stress-strain data for each of them. With this data reduction, several properties will be compared between the three designs such as axial stiffness, yield stress, and fracture load. Comparing the stress-strain data for each spar configuration is important to drawing conclusions about the types of spar cross-sections that perform the best under appropriate loading.

Experimental Setup

Specimen Designs

There are three spar designs to be tested. Each design—one with longitudinal bars, one with X-shaped reinforcements, and the other with transverse bars—is shown in Figure 1. Figures 3(a), 3(b), and 4 show the part drawings for each of the spar sections. They were designed to be reflective of spar sections that may be seen on existing aerodynamic structures and structures that appear in other engineering applications that prove to be the most weight-bearing. The specimens were also designed to be able to install onto the Instron machine for the experiments described in

the following section.



Figure 1: The specimens with longitudinal bars, the X-shaped reinforcements, and transverse bars from top to bottom.

Testing Procedure

Figure 11 shows how the specimens are installed within the arms of the Instron machine. Each test involved the specimen to be loaded in compression until failure. The loading occurs when the arm is lowered into the spar.

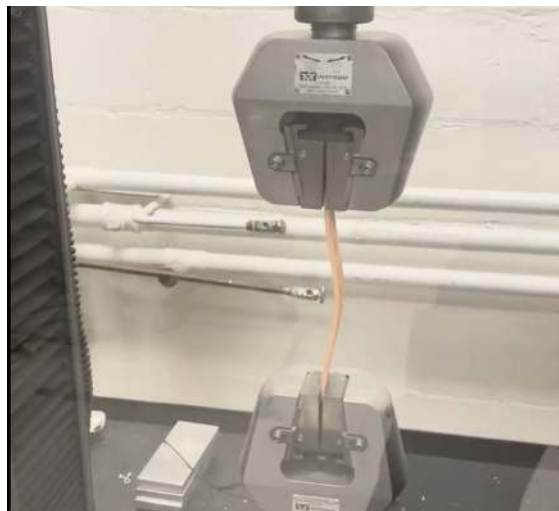
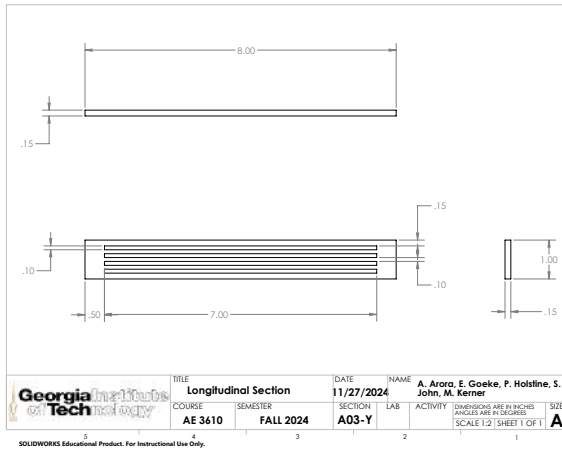
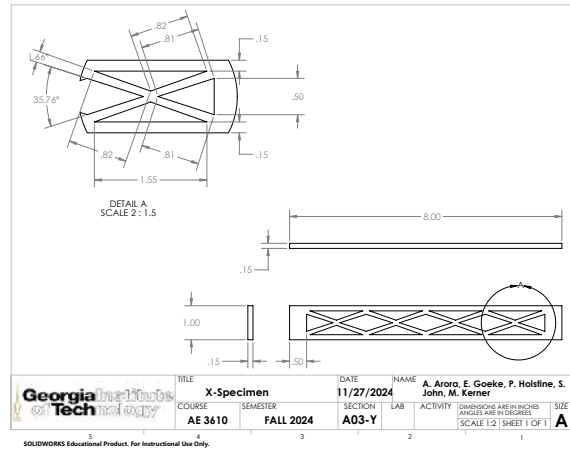


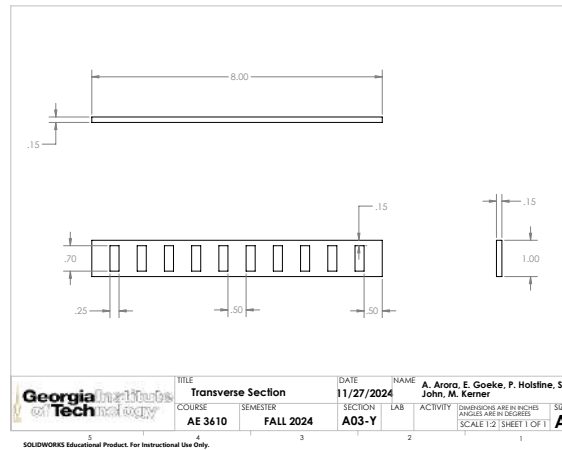
Figure 2: The Instron tensile test machine during a compression test with a spar.



(a) Longitudinal specimen drawing



(b) X specimen drawing



(c) Transverse specimen drawing

Figure 3: Solidworks part drawings for each specimen.

Results and Discussion

Raw Data

The first part of the data for this report is the working length and cross sectional areas of each specimen. These values are found in Tables I and II and are necessary for stress and strain calculations.

The next part of the data is the observed buckling load values found during the experiment for each specimen. These can be split into local and global bucklings, although the vertical column specimen is the only one that had observed local buckling. These values are also found in

Table I.

Table I: Working lengths and buckling loads (local and global) for each specimen

Specimen	Working Length (cm)	Local Buckling Load (N)	Global Buckling Load (N)
Horizontal	17.6	N/A	121
Vertical	17.6	82	131
"X" Specimen	17.7	N/A	106

Table II: Cross-sectional areas

Measurements	Values (in²)
Hole Section (Horizontal Bar)	0.045
Infilled (Horizontal Bar)	0.15
Column Area (Vertical Column)	0.09
Middle Diamond ("X" Specimen)	0.0623
Lower Diamond ("X" Specimen)	0.06825

The rest of the raw data in this report comes straight from the Instron machine used for testing. The Instron can record and output data about both force and displacement and in this case, force in compression. The units for the raw data are Newtons for force and millimeters for displacement, and it is important to mention that the same procedure was done for each of the beams. The data was recorded by a camera and then was placed into a CSV file later with our team selecting different points (all with roughly the same interval of time in between) for each specimen. This data can be found in Tables III, IV, and V, for the horizontal beam specimen, the vertical beam specimen, and the diagonal beam specimen respectively.

Table III: Table for values used in the report for the horizontal beam specimen

Displacement (mm)	Force (N)	Strain	Stress (filled section)	Stress (open section)
0.002	0.5243	0.000113636	5.418001447	18.06062694
0.05435	19.759	0.003088068	204.1851814	680.6407165
0.08131	25.769	0.004619886	266.291206	887.6679297
0.1117	41.163	0.006346591	425.3694327	1417.946951
0.16195	53.341	0.009201705	551.2142193	1837.444023
0.19023	66.181	0.010808523	683.899969	2279.745091
0.21787	78.086	0.012378977	806.9236334	2689.838099
0.24805	84.836	0.01409375	876.676656	2922.356183
0.30929	94.311	0.017573295	974.5892322	3248.74268
0.33372	100.1	0.018961364	1034.411491	3448.157079
0.4784	105.9	0.027181818	1094.347422	3647.950396
0.58153	106.2	0.033041477	1097.447556	3658.284533
0.62524	111.5	0.035525	1152.216596	3840.854289
0.7034	112.36	0.039965909	1161.103648	3870.478815
0.74361	113.04	0.042250568	1168.130619	3893.902859
0.77411	113.83	0.043983523	1176.294306	3921.116087
0.79677	114.23	0.045271023	1180.427819	3934.894936
0.83695	115.05	0.047553977	1188.901519	3963.141578
0.87367	115.84	0.049640341	1197.065206	3990.354805
0.89789	116.24	0.051016477	1201.198719	4004.133655
0.92178	116.93	0.052373864	1208.329028	4027.90217
0.96122	117.59	0.054614773	1215.149323	4050.637272
1.0085	118.12	0.057301136	1220.626227	4068.894247
1.046	118.58	0.059431818	1225.379766	4084.739924
1.0824	119.2	0.0615	1231.786711	4106.097141
1.1261	119.51	0.063982955	1234.990183	4116.775749
1.1537	119.67	0.065551136	1236.643588	4122.287289
1.1912	119.94	0.067681818	1239.433709	4131.588012
1.2092	120.2	0.068704545	1242.120492	4140.544265
1.2328	120.31	0.070045455	1243.257208	4144.333448
1.2588	120.41	0.071522727	1244.290586	4147.778161
1.3083	120.62	0.074335227	1246.46068	4155.012056
1.3398	120.85	0.076125	1248.83745	4162.934895
1.3769	120.98	0.078232955	1250.180841	4167.413021
1.423	121.08	0.080852273	1251.214219	4170.857733
1.4656	121.17	0.083272727	1252.14426	4173.957975
1.4984	121.22	0.085136364	1252.660949	4175.680331
1.5745	121.28	0.089460227	1253.280976	4177.747158
1.7101	121.33	0.097164773	1253.797665	4179.469514
1.8449	121.35	0.104823864	1254.00434	4180.158457
1.927	121.29	0.109488636	1253.384313	4178.091629
2.0269	121.19	0.115164773	1252.350935	4174.646917
2.0892	121.15	0.118704545	1251.937584	4173.269032
2.1926	120.99	0.124579545	1250.284179	4167.757492
2.2842	120.87	0.129784091	1249.044125	4163.623837
2.3758	120.7	0.134988636	1247.287382	4157.767826
2.5904	120.33	0.147181818	1243.463883	4145.022391
2.7142	120.02	0.154215909	1240.260411	4134.343782
2.9415	119.47	0.167130682	1234.576832	4115.397864

Table IV: Table for values used in the report for the vertical beam specimen

Displacement (mm)	Force (N)	Strain	Stress
0.01199	0.40495	0.00068125	6.97588286
0.03748	8.6258	0.002129545	148.5925926
0.04149	23.063	0.002357386	397.295435
0.05348	30.774	0.003038636	530.129199
0.07326	48.216	0.0041625	830.5943152
0.086	53.75	0.004886364	925.9259259
0.10751	58.863	0.006108523	1014.005168
0.14626	59.15	0.008310227	1018.949182
0.16104	59.473	0.00915	1024.513351
0.16555	65.296	0.00940625	1124.823428
0.18172	70.473	0.010325	1214.005168
0.20976	77.184	0.011918182	1329.612403
0.2609	82.758	0.014823864	1425.633075
0.2965	82.651	0.016846591	1423.789836
0.33621	82.579	0.019102841	1422.549526
0.38189	82.574	0.021698295	1422.463394
0.4	82.538	0.022727273	1421.843239
0.43323	82.928	0.024615341	1428.561585
0.44628	83.222	0.025356818	1433.626184
0.46546	93.015	0.026446591	1602.325581
0.50318	98.221	0.028589773	1692.006891
0.52619	106.62	0.029897159	1836.692506
0.55108	107.23	0.031311364	1847.200689
0.5652	113.58	0.032113636	1956.589147
0.58133	114.03	0.033030114	1964.341085
0.60379	119.65	0.03430625	2061.154177
0.65626	120.29	0.0372875	2072.179156
0.67385	125.84	0.038286932	2167.786391
0.72156	126.01	0.040997727	2170.714901
0.76655	126.57	0.043553977	2180.361757
0.79803	127.1	0.045342614	2189.491817
0.82833	127.6	0.047064205	2198.105082
0.87886	128.08	0.049935227	2206.373816
0.9268	129.05	0.052659091	2223.083549
0.95311	129.25	0.054153977	2226.528854
0.9883	129.61	0.056153409	2232.730405
1.0074	129.8	0.057238636	2236.003445
1.0556	130.3	0.059977273	2244.61671
1.0971	130.66	0.062335227	2250.81826
1.1161	130.77	0.063414773	2252.713178
1.1435	130.86	0.064971591	2254.263566
1.1845	131.09	0.067301136	2258.225668
1.215	131.2	0.069034091	2260.120586
1.2452	131.22	0.07075	2260.465116
1.2697	131.2	0.072142045	2260.120586
1.2944	131.14	0.073545455	2259.086994
1.3741	131.09	0.078073864	2258.225668
1.4557	131.05	0.082710227	2257.536606
1.5068	131.02	0.085613636	2257.019811

Table V: Table for values used in the report for the diagonal beam specimen

Displacement (mm)	Force (N)	Strain	"X" Section	"Diamond" Section
0	0.4	0	9.185203556	8.2666832
0.0246	10.2	0.001397727	234.2226907	210.8004216
0.0418	17.8	0.002375	408.7415582	367.8674024
0.0644	25.6	0.003659091	587.8530276	529.0677248
0.0891	39.9	0.0050625	916.2240547	824.6016492
0.117	46.4	0.006647727	1065.483612	958.9352512
0.1474	57.2	0.008375	1313.484108	1182.135698
0.1821	69.4	0.010346591	1593.632817	1434.269535
0.2312	74.6	0.013136364	1713.040463	1541.736417
0.2453	81.6	0.0139375	1873.781525	1686.403373
0.2682	91.4	0.015238636	2098.819012	1888.937111
0.3067	98.3	0.017426136	2257.263774	2031.537396
0.3334	103.5	0.018943182	2376.67142	2139.004278
0.3821	109.4	0.021710227	2512.153172	2260.937855
0.4352	110.1	0.024727273	2528.227279	2275.404551
0.469	117.1	0.026647727	2688.968341	2420.071507
0.524	117.4	0.029772727	2695.857244	2426.271519
0.585	118.3	0.033238636	2716.523952	2444.871556
0.6404	123.4	0.036386364	2833.635297	2550.271767
0.7036	123.7	0.039977273	2840.5242	2556.47178
0.7314	124	0.041556818	2847.413102	2562.671792
0.7925	124.9	0.045028409	2868.07981	2581.271829
0.8264	125.5	0.046954545	2881.857616	2593.671854
0.8637	125.9	0.049073864	2891.042819	2601.938537
0.9013	126.2	0.051210227	2897.931722	2608.13855
0.9403	126.9	0.053426136	2914.005828	2622.605245
0.959	127.3	0.054488636	2923.191032	2630.871928
1.014	127.6	0.057613636	2930.079934	2637.071941
1.047	127.8	0.059488636	2934.672536	2641.205282
1.09	128.3	0.061931818	2946.15404	2651.538636
1.129	128.4	0.064147727	2948.450341	2653.605307
1.169	128.7	0.066420455	2955.339244	2659.80532
1.193	128.9	0.067784091	2959.931846	2663.938661
1.24	129.1	0.070454545	2964.524448	2668.072003
1.275	129.3	0.072443182	2969.117049	2672.205344
1.316	129.6	0.074772727	2976.005952	2678.405357
1.336	129.8	0.075909091	2980.598554	2682.538698
1.365	129.9	0.077556818	2982.894855	2684.605369
1.421	130	0.080738636	2985.191156	2686.67204
1.459	130.2	0.082897727	2989.783757	2690.805382
1.494	130.2	0.084886364	2989.783757	2690.805382
1.545	130.3	0.087784091	2992.080058	2692.872052
1.593	130.4	0.090511364	2994.376359	2694.938723
1.673	130.4	0.095056818	2994.376359	2694.938723
1.754	130.3	0.099659091	2992.080058	2692.872052
1.819	130.3	0.103352273	2992.080058	2692.872052
1.859	130.2	0.105625	2989.783757	2690.805382
1.914	130	0.10875	2985.191156	2686.67204
1.968	129.1	0.111818182	2964.524448	2668.072003

Data Reduction

The data reduction is done the same way for each of the test specimens. The first plot for the specimen is a force vs. displacement plot. The plot is useful to see the buckling of the beam, and at what point it gives out. These plots are shown in Figures 4, 5, and 6, for the horizontal bar specimen, vertical bar specimen, and diagonal bar specimen respectively.

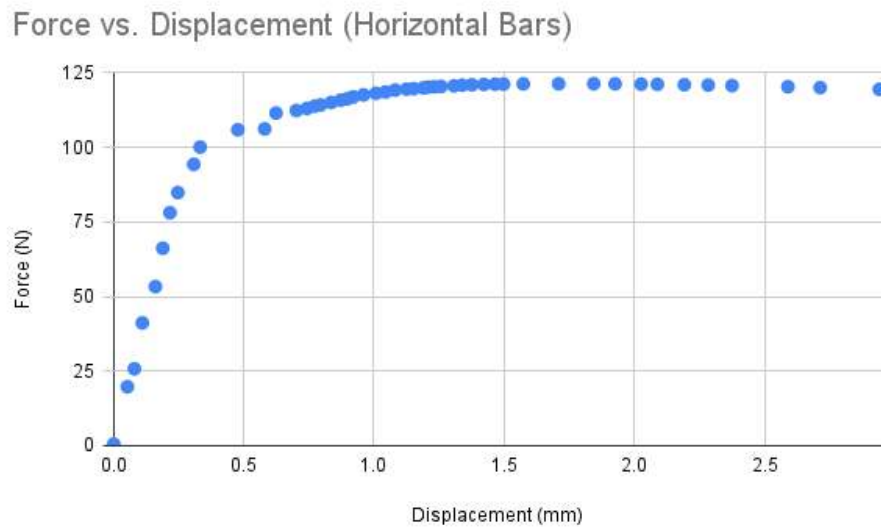


Figure 4: Force vs. displacement for the specimen with the horizontal bars

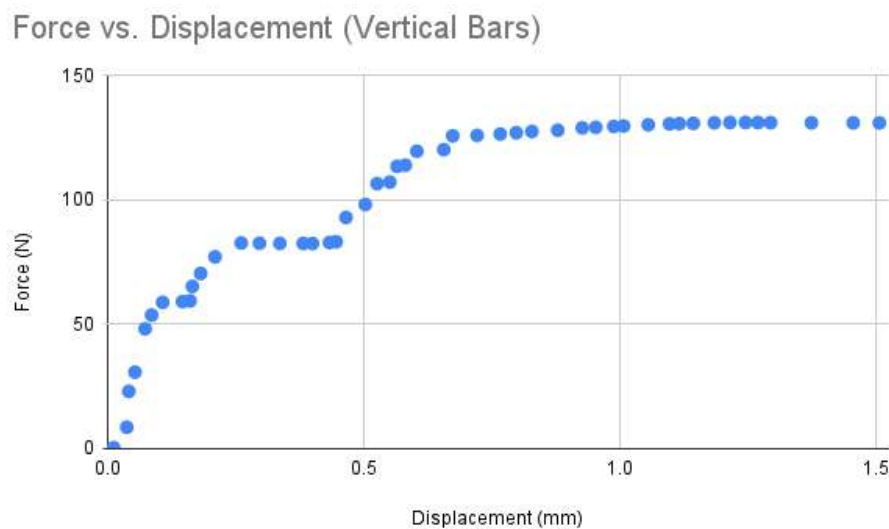


Figure 5: Force vs. displacement for the specimen with the vertical bars

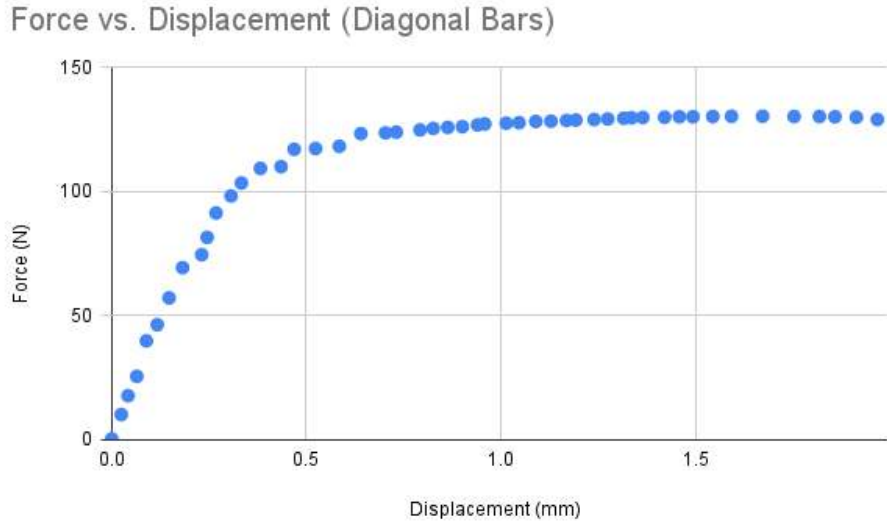


Figure 6: Force vs. displacement for the specimen with the diagonal bars

Next, the strain values were calculated using Equation (1).

$$\epsilon = \frac{L - L_0}{L_0} = \frac{\delta}{L_0} \quad (1)$$

The δ value is the change in length, which is simply the displacement value recorded by the Instron. The L_0 value is the initial length of the beam which is shown in the tables for initial measurements of the specimen, namely Tables I, and II.

The resulting plots from these calculated strain values are the force vs. strain plots which are shown in Figures 7, 8, and 9 for the horizontal beam, vertical beam, and diagonal beam respectively. these plots are similar to the force vs. displacement graph, this makes sense however since the strain is dependent upon the ratio of the displacement and the length.

Force vs. Strain (Horizontal Bars)

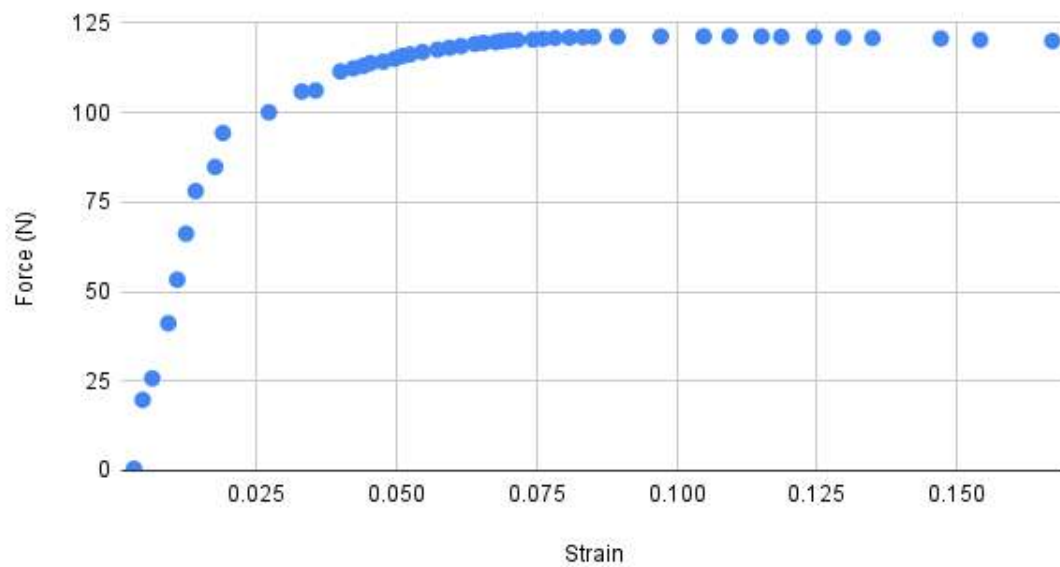


Figure 7: Force vs. strain for the horizontal bar specimen

Force vs. Strain (Vertical Bars)

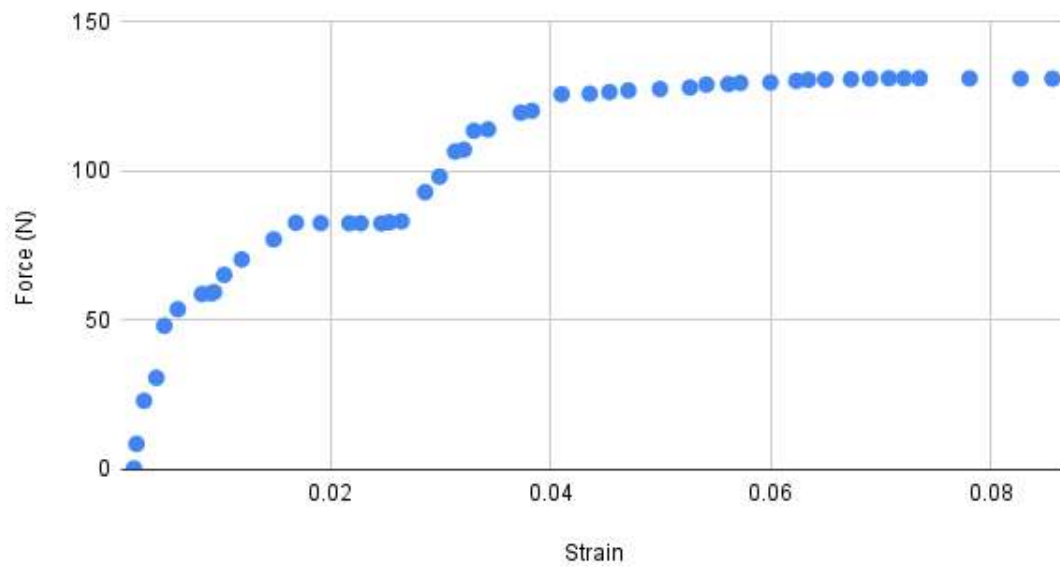


Figure 8: Force vs. strain for the vertical bar specimen

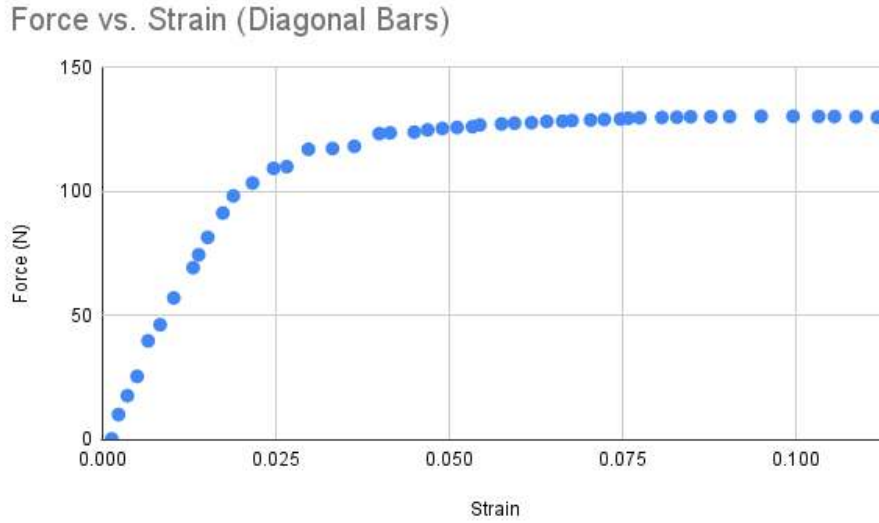


Figure 9: Force vs. strain for the diagonal bar specimen

Moving on from strain, we can calculate the stress values through the use of Equation (2).

$$\sigma = \frac{P}{A} \quad (2)$$

In this equation, P is the value for the load applied, which was negative from the raw data due to compression, and A is the value for the cross-sectional area. The magnitude of load values were used to make the strain values positive. These are values we have using Tables I and II alongside the recorded Instron force values. This of course means that there will be different values for stress for the beams that have holes in them, there will be stress for the open section as well as the filled section for each of these plots. However, the area values for both the filled and open sections are recorded in the aforementioned tables.

The plots that result from the calculated stress values are placed in a stress vs. strain plot, with different lines representing the open sections on some of the beams. The plots are shown for the horizontal beam specimen, vertical beam specimen, and diagonal beam specimen in Figures 10, 11, 12 respectively. The value for stress is in kPa, meaning that the value for the original stress which is in $\frac{N}{m^2}$ must be divided by 1000, or multiplied by $\frac{1}{1000}$ to get the stress in kPa.

In total, the load and displacement values recorded for each specimen were first organized, then used to solve for stress and strain. Then, with all of this, plots of force vs displacement, stress vs strain, and force vs strain were created for all relevant specimens and points within these specimens.



Figure 10: Stress vs. strain for the horizontal bar specimen

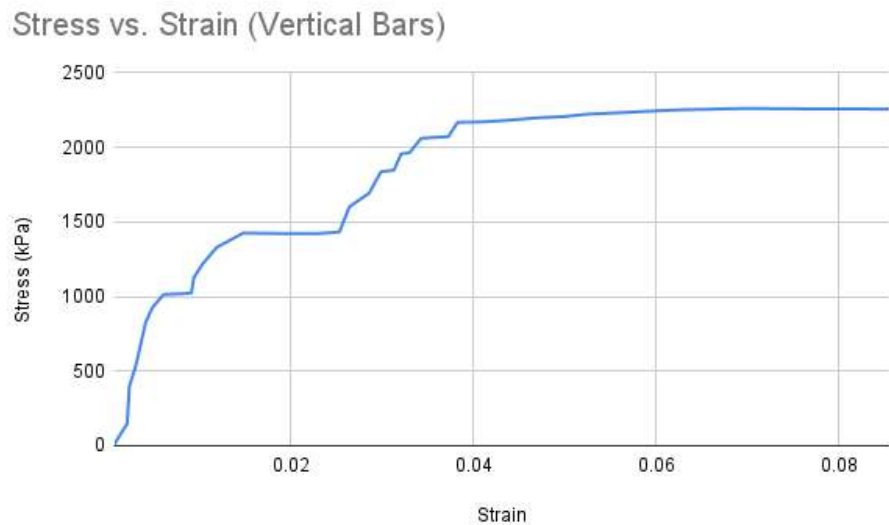


Figure 11: Stress vs. strain for the vertical bar specimen

Stress vs Strain (Diagonal Beams)

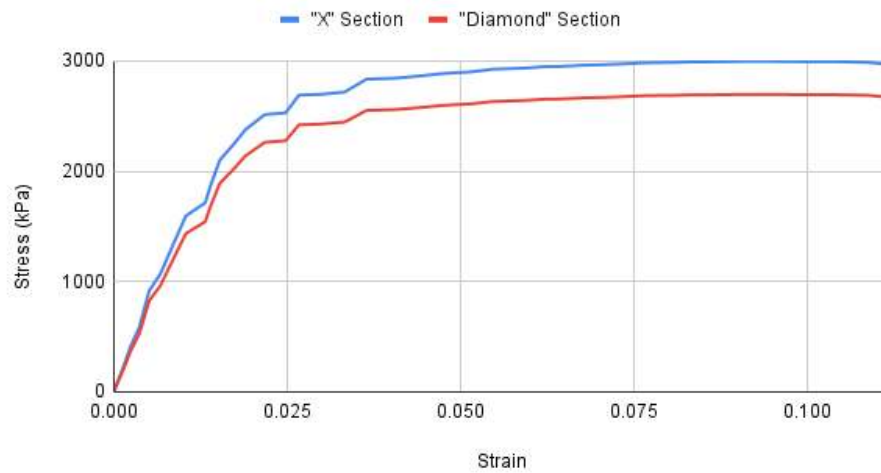


Figure 12: Stress vs. strain for the diagonal bar specimen

The axial stiffness of each frame can be found by taking the average slope of the elastic region of their stress-strain plots. Using the data from these experiments, Table VI shows the axial stiffnesses, yield stresses, and fracture loads deduced for each configuration.

Cross-section		Effective Stiffness (N/m)	Yield Stress (kPa)	Fracture Load (N)
Horizontal Bars	Open Section	383.1063269	3448	119.5
	Filled Section		876.7	
Vertical Bars		791.7217501		131.02
Diagonal Bars	X-section	268.3823529	1594	129.1
	Diamond Section		1434	

Table VI: Comparison of Cross-Section Properties

Discussion

This experiment analyzed three simple cross-sectional patterns to determine which offered the best structural integrity under compression. By identifying when and where each specimen buckled, conclusions were drawn about optimal cross-section designs. The vertical bar specimen performed the worst, experiencing local buckling at 82 N and global buckling at 131 N. The early

local buckling was likely due to its slender columns, offering minimal support.

The "X" specimen demonstrated moderate performance, buckling under a compressive load of 106 N. Despite being a hybrid design, its strength was lower than expected, likely due to the long, slender bars within its cross-section. A denser arrangement of smaller Xs could improve its structural integrity, as shorter bars provide better support.

The horizontal bar specimen performed best, buckling at 121 N. Its superior strength can be attributed to the higher material density within the cross-section, which provided greater resistance to buckling.

Figure 5 highlights the behavior of the vertical bars under force, showing an initial plateau where local buckling occurred before displacement increased and global buckling followed.

Stress and strain analysis further reinforced the importance of minimizing open space in cross-sections. While a fully solid design is impractical due to cost and weight, balancing material distribution and structural integrity is critical. The horizontal bar's performance aligns with this principle, and a modified "X" specimen with smaller, denser Xs would likely outperform all tested designs.

Summary and Conclusions

This study investigated the structural performance of three cross-sectional aircraft spar patterns: horizontal bars, vertical bars, and X-shaped designs—under axial compression. To conclude, this experiment displayed some expected results and some surprising revelations. In order to truly maximize the cross section, one could continue to test the hypothesis drawn from this experiment. By maximizing the ratio of strength to material weight within, many structures in aeronautics such as wings or landing parts could be improved. If aircraft are maximized for efficiency and become lighter and stronger, they can also move faster, and mankind can push past previous limits.

The horizontal bar cross-section proved to be the most structurally efficient design under compression, highlighting the importance of material density in resisting buckling. Vertical bar

specimens were prone to early failure due to local buckling, making them less suitable for load-bearing applications. While the X-shaped design did not outperform the horizontal bars, its potential for optimization suggests it could become a superior alternative with modifications such as reducing the length of its slender members or increasing the number of Xs within the cross-section.

These findings provide valuable insights into the optimization of beam cross-section designs for applications requiring high load-bearing capacity, such as in aeronautical engineering. Future work should explore the effects of material properties, manufacturing methods, and geometric modifications on the performance of X-shaped and other novel cross-sectional patterns.